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PROJECT REPORT

Course Code: CSA2001

Spam Detection Using Machine Learning

A Comparative Study of Classification Algorithms

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1. Introduction

Spam messages pose significant challenges in digital communication — wasting user time, distributing malware, and cluttering important channels. With over 300 billion emails sent daily and SMS spam rising 30% annually, automated machine learning detection is essential. This project compares three classical ML classifiers — Naive Bayes, Logistic Regression, and Random Forest — on the SMS Spam Collection dataset.

2. Dataset & Methodology

The UCI SMS Spam Collection dataset (5,574 messages: 86.6% ham, 13.4% spam) was used. Messages were preprocessed — lowercased, cleaned, and tokenised. TF-IDF vectorisation with 3,000 features was applied, with an 80/20 stratified train-test split.

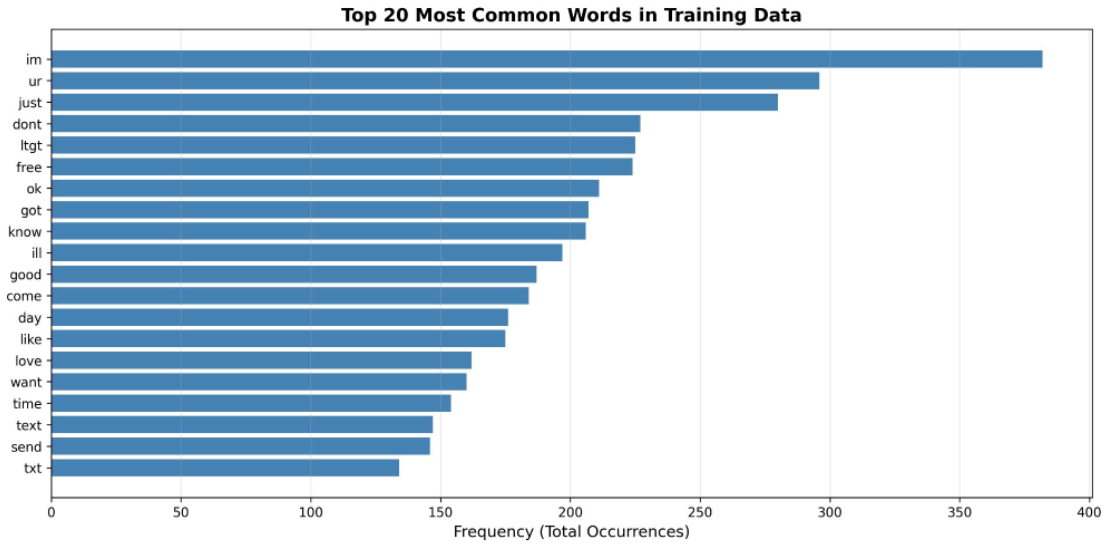

Figure 1: Top 20 Most Common Words in Training Data

3. Results & Analysis

All three algorithms exceeded 95% accuracy. The table and figures below summarise the complete evaluation.

Algorithm	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Naive Bayes	97.13%	100.00%	78.52%	87.97%	0.9817
Logistic Regression	96.59%	100.00%	74.50%	85.38%	0.9827
Random Forest (Best)	**97.85%**	**100.00%**	**83.89%**	**91.24%**	**0.9801**

Table 1: Performance Metrics Comparison

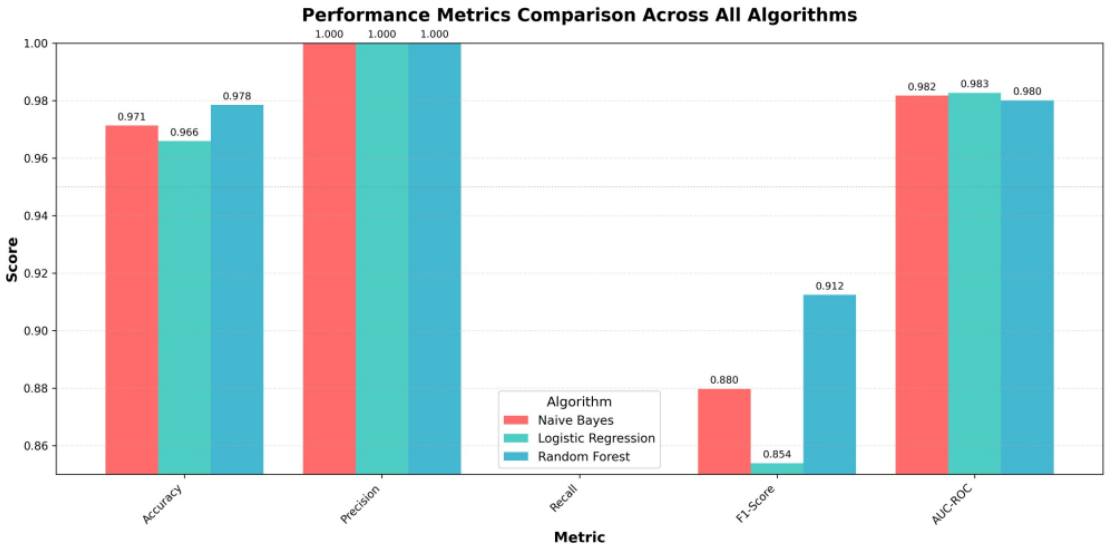

Figure 2: Performance Metrics Comparison Across All Algorithms

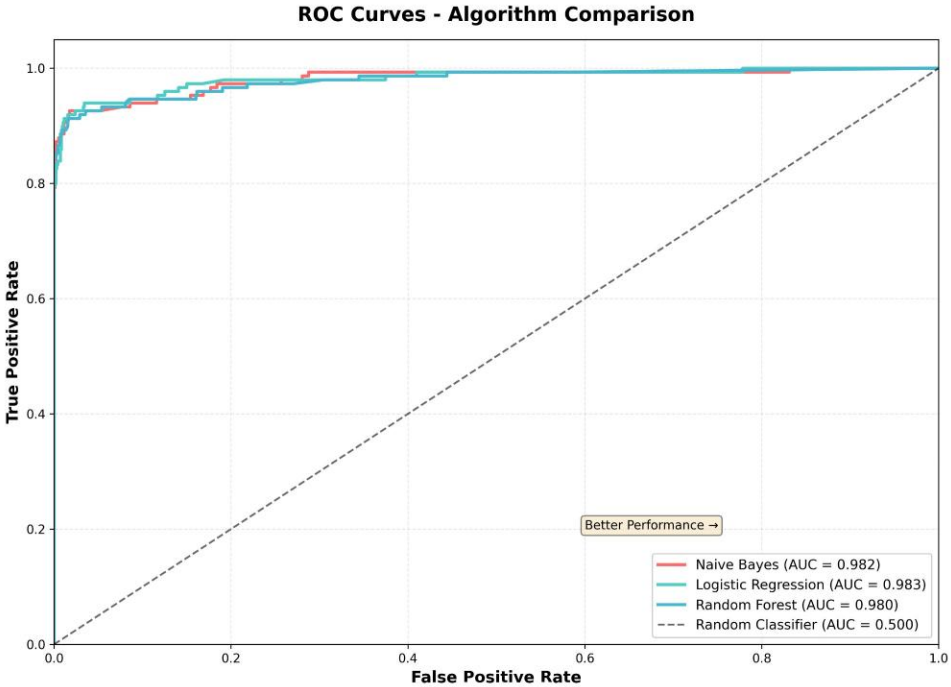

Figure 3: ROC Curves — AUC Comparison

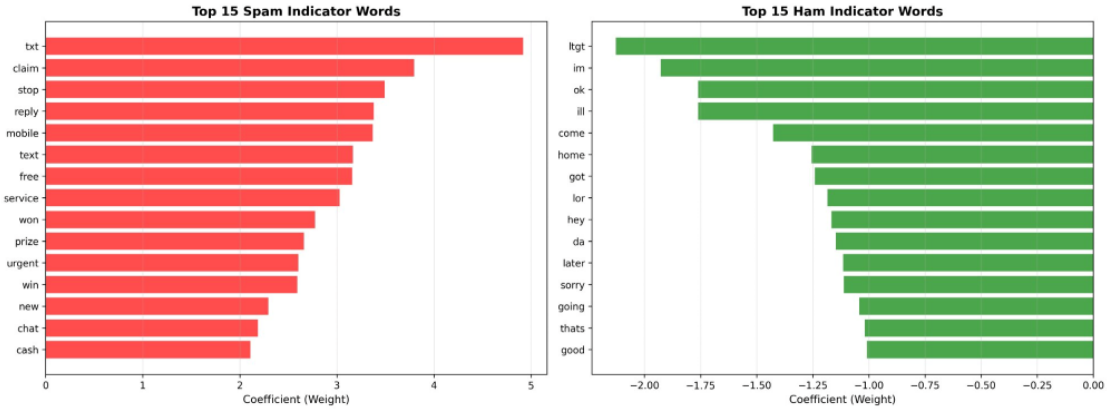

Figure 4: Top Spam & Ham Indicator Words (Logistic Regression)

4. Key Findings

- Random Forest: Best accuracy (97.85%) and F1-Score (91.24%) — recommended for production.
- Naive Bayes: Best speed-accuracy balance — ideal for real-time filtering.
- All models: 100% Precision — zero legitimate messages flagged as spam.
- All AUC-ROC scores above 0.98 — excellent discriminative performance.
- Top spam keywords: 'free', 'txt', 'claim', 'prize', 'urgent', 'win'.

5. Conclusion

Classical machine learning algorithms are highly effective for spam detection. Random Forest is recommended for accuracy-critical deployments; Naive Bayes suits real-time systems. Future work could explore BERT or LSTM-based deep learning models for further improvement.